K J SOMAIYA INSTITUTE OF TECHNOLOGY

Department of Information Technology

Academic Year: 2026-27
Class: S.Y. B.Tech.
Semester: III

Subject: DSA LAB
Course Code: ITL302

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Division: IT-A
Batch: SA2
Roll Number: 25

Experiment No:5

Aim: Implementation of Singly Linked List and various operations for real-world.

Program:

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int temperature;
    struct Node *next;
};

struct Node *head = NULL;

void insertAtStart(int temp) {
    struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
    newNode->temperature = temp;
    newNode->next = head;
    head = newNode;

    printf("Inserted temperature: %dC\n", temp);

    if (temp >= 45.0){
        printf(" ALERT: Severe Heatwave Warning! Temperature %dC is dangerously
high.\n", temp);
    }
    else if (temp >= 40.0){
        printf(" WARNING: Heatwave conditions detected (%dC).\n", temp);
    }
}

void deleteAtStart() {
    if (head == NULL) {
        printf("List is empty. No temperature reading to delete.\n");
        return;
    }
    struct Node *temp = head;
    head = head->next;
    printf("Deleted temperature reading: %dC\n", temp->temperature);
```

```
    free(temp);
}

void display() {
    if (head == NULL) {
        printf("No temperature observations recorded.\n");
        return;
    }
    struct Node *current = head;
    int count = 1;
    printf("\n--- Temperature Observations (most recent first) ---\n");
    while (current != NULL) {
        printf("%d. %dC", count, current->temperature);
        if (current->temperature >= 45.0)
            printf(" [SEVERE HEATWAVE]");
        else if (current->temperature >= 40.0)
            printf(" [HEATWAVE WARNING]");
        printf("\n");
        current = current->next;
        count++;
    }
    printf("-------------------------------------------------\n");
}

void freeList() {
    struct Node *current = head;
    struct Node *next;
    while (current != NULL) {
        next = current->next;
        free(current);
        current = next;
    }
    head = NULL;
}

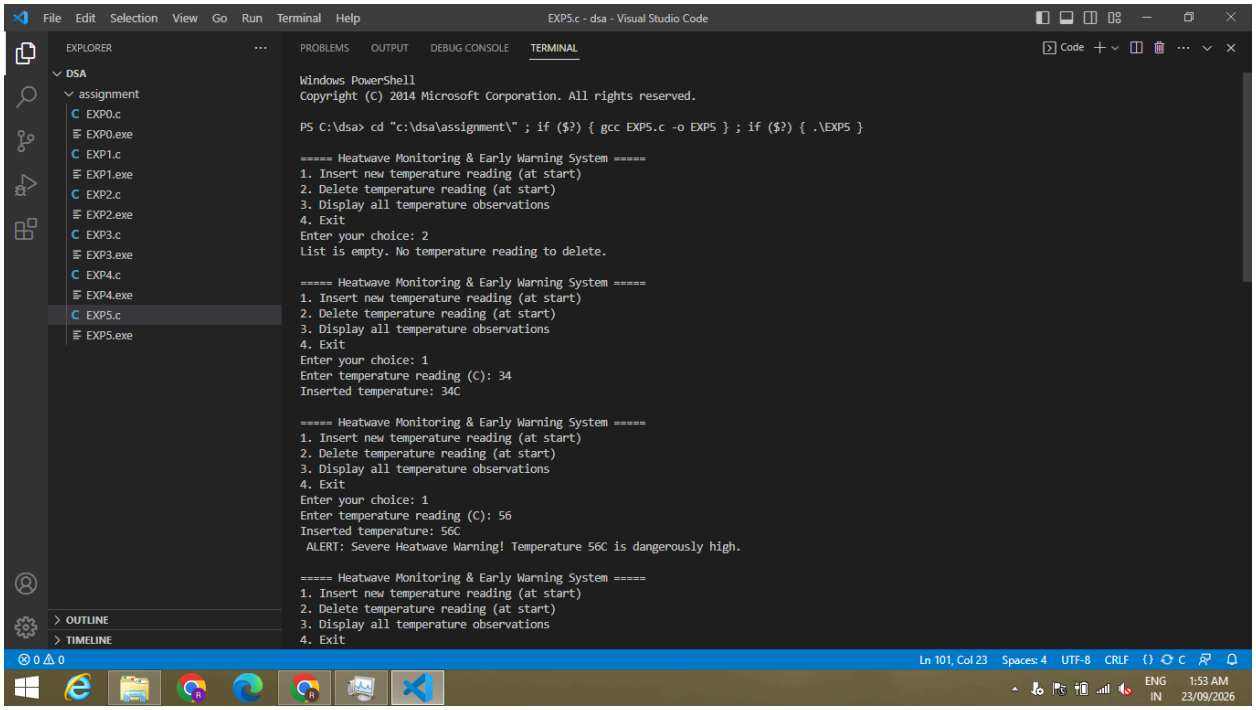
int main() {
    int choice;
    int temp;

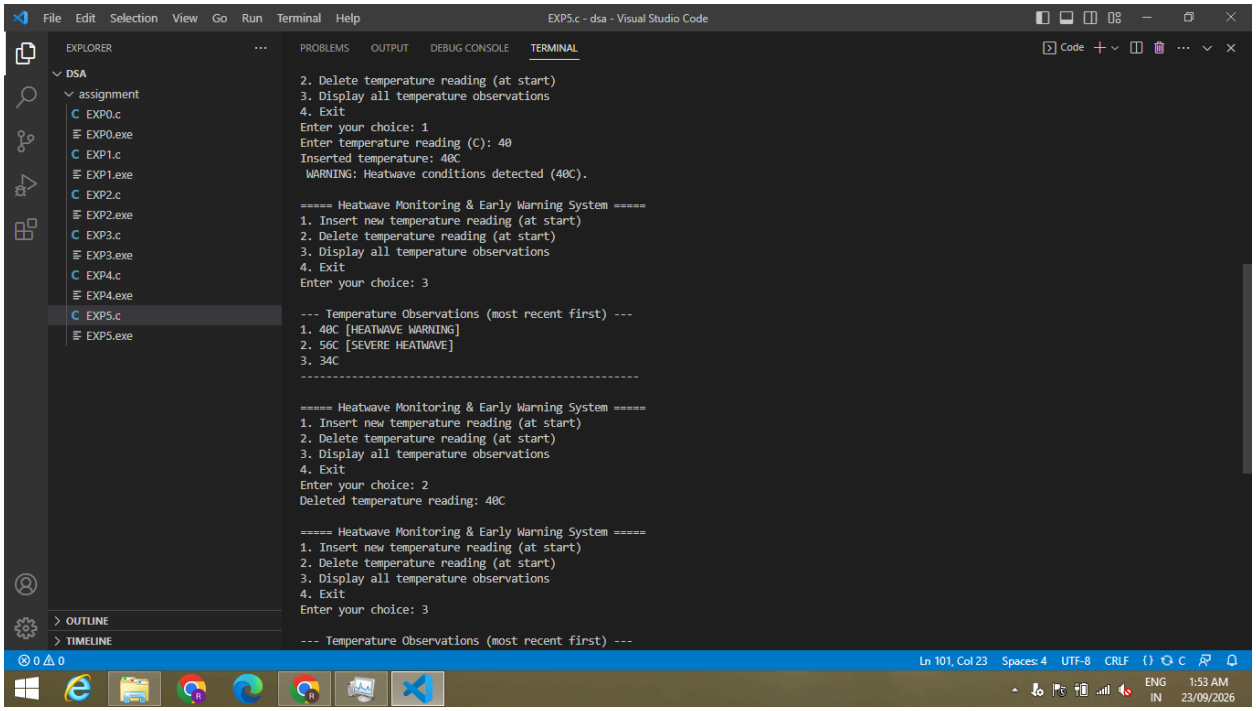
    do {
```

```
        printf("\n===== Heatwave Monitoring & Early Warning System =====\n");
        printf("1. Insert new temperature reading (at start)\n");
        printf("2. Delete temperature reading (at start)\n");
        printf("3. Display all temperature observations\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

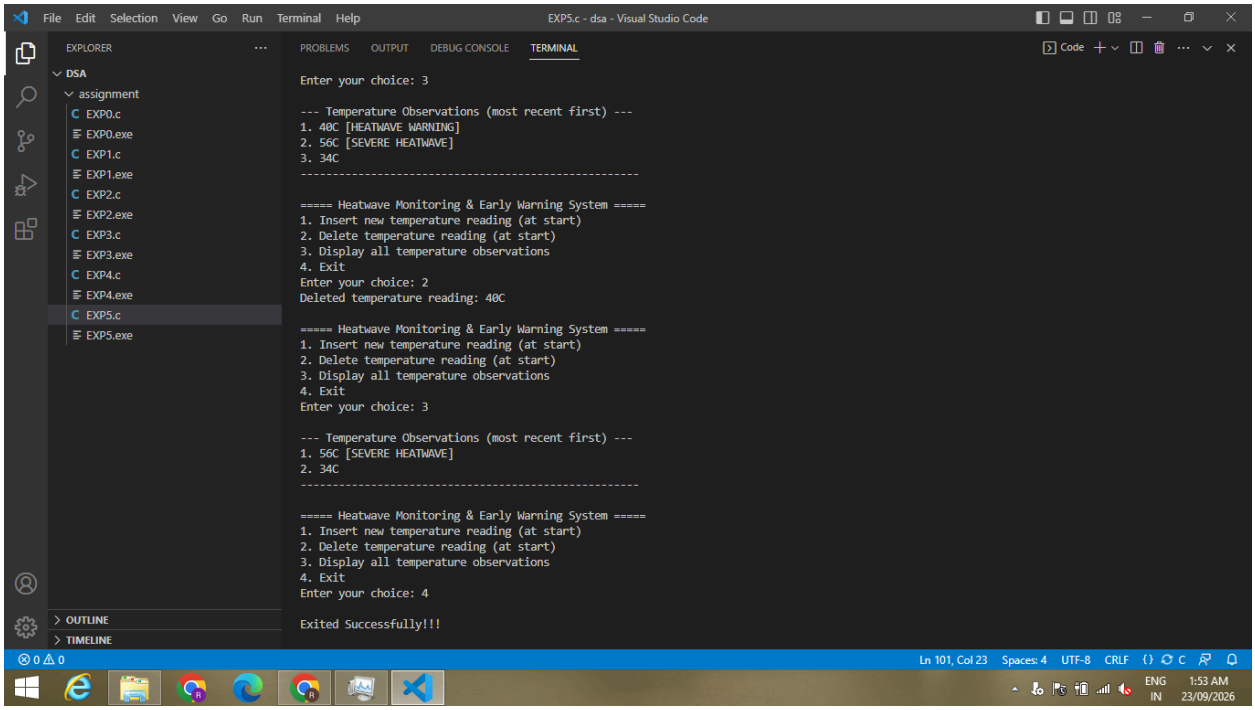
        switch (choice) {
            case 1:
                printf("Enter temperature reading (C): ");
                scanf("%d", &temp);
                insertAtStart(temp);
                break;
            case 2:
                deleteAtStart();
                break;
            case 3:
                display();
                break;
            case 4:
                printf("\nExited Successfully!!!\n");
                freeList();
                break;
            default:
                printf("\nInvalid Input!\nEnter betweem 1 to 4\n");
                break;
        }
    }while(choice!=4);
    return 0;
}
```

Outcome:


File Edit Selection View Go Run Terminal Help
EXP5.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
EXP0.c
EXP0.exe
EXP1.c
EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
EXP5.exe
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
Windows PowerShell
Copyright (C) 2014 Microsoft Corporation. All rights reserved.
PS C:\dsa> cd "c:\dsa\assignment\" ; if ($?) { gcc EXP5.c -o EXP5 } ; if ($?) { .\EXP5 }
===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 2
List is empty. No temperature reading to delete.
===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 1
Enter temperature reading (C): 34
Inserted temperature: 34C
===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 1
Enter temperature reading (C): 56
Inserted temperature: 56C
ALERT: Severe Heatwave Warning! Temperature 56C is dangerously high.
===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Ln 101, Col 23 Spaces: 4 UTF-8 CRLF C
ENG IN 1:53 AM 23/09/2026


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EXP5.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
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EXP0.exe
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EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
EXP5.exe
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 1
Enter temperature reading (C): 40
Inserted temperature: 40C
WARNING: Heatwave conditions detected (40C).
===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 3
--- Temperature Observations (most recent first) ---
1. 40C [HEATWAVE WARNING]
2. 56C [SEVERE HEATWAVE]
3. 34C

===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 2
Deleted temperature reading: 40C
===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 3
--- Temperature Observations (most recent first) ---
Ln 101, Col 23 Spaces: 4 UTF-8 CRLF C
ENG IN 1:53 AM 23/09/2026


File Edit Selection View Go Run Terminal Help
EXP5.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
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EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
EXP5.exe
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
Enter your choice: 3
--- Temperature Observations (most recent first) ---
1. 40C [HEATWAVE WARNING]
2. 56C [SEVERE HEATWAVE]
3. 34C

===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 2
Deleted temperature reading: 40C
===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 3
--- Temperature Observations (most recent first) ---
1. 56C [SEVERE HEATWAVE]
2. 34C

===== Heatwave Monitoring & Early Warning System =====
1. Insert new temperature reading (at start)
2. Delete temperature reading (at start)
3. Display all temperature observations
4. Exit
Enter your choice: 4
Exited Successfully!!!
0 0
Ln 101, Col 23 Spaces: 4 UTF-8 CRLF C
ENG IN
1:53 AM 23/09/2026